Education City High School

2.2

Change in Linear and Exponential Functions

Student Notes and Guided Practice

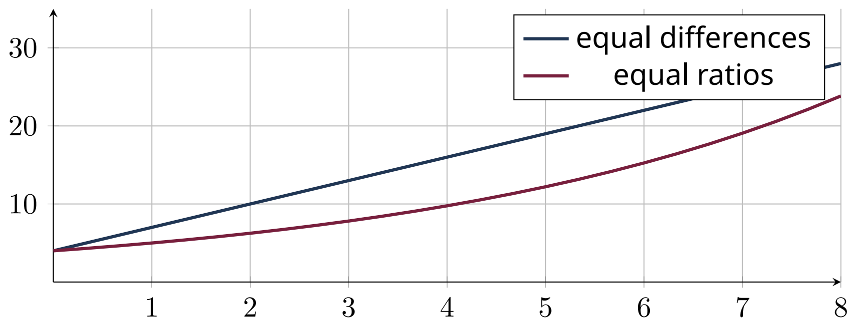
AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Learning Targets

- Compare additive and proportional change.
- Construct linear and exponential functions from values or rates.
- Interpret behavior on discrete and continuous domains.

Essential Question

How do linear and exponential functions encode fundamentally different rate patterns?



Concept Framework

Key Idea

Linear functions add a constant amount over equal intervals.

Key Idea

Exponential functions multiply by a constant factor over equal intervals.

Key Idea

A percent rate p corresponds to factor $1 + p$ for growth and $1 - p$ for decay.

Key Idea

Two points determine either model once the family is specified.

Formula Map

Formula and Structure

$$L(x) = L_0 + mx$$

Formula and Structure

$$E(x) = E_0 b^x$$

Formula and Structure

$$b = \left(\frac{E(x_2)}{E(x_1)}\right)^{1/(x_2 - x_1)}$$

Worked Examples

Worked Example 1

Problem. Construct a linear function with $L(2) = 11$ and $L(7) = 31$.
Solution. $m = 4$, so $L(x) = 4x + 3$.

Worked Example 2

Problem. Construct an exponential function with $E(0) = 6$ and $E(3) = 48$.
Solution. $b^3 = 8$, so $b = 2$; $E(x) = 6 \cdot 2^x$.

Worked Example 3

Problem. Compare a 5% yearly increase with adding 5 units yearly.
Solution. The 5% increase is proportional to the current amount, so the absolute change grows; adding 5 gives constant absolute change.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Construct the linear function through (0, 5) and (4, 17).

2. Construct the exponential function through (0, 5) and (4, 80).

3. Construct the linear function through (2, 10) and (7, 35).

4. Construct the exponential function through (1, 6) and (3, 54).

5. Construct the linear function through (0, 12) and (5, 6).

6. A value increases from 40 to 52 in 3 years. Compare the linear yearly change and exponential yearly factor.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is ____________________.
2. A restriction I must remember is ____________________.
3. One graphical feature I can justify algebraically is ____________________.

Video support: Flipped Math topic page | Original ECHS material informed by Pearson and Holt Precalculus concepts.